Institutional Research Platform — Plan 1 of 3: Foundations

Repo: `~/Workspace/institutional-research` · **Author:** Claude (Opus 5) for Markandeya Varma
Date: 2026-08-16 · **Status:** Phase 0 deliverable — awaiting owner approval before any code
Predecessor: MICCV2 at `~/Workspace/MICCV2`, to be frozen read-only

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0. What this document is, and what it is not

This is the Phase 0 audit and architecture specification that the owner's plan §25 requires *before* production code is written. It covers foundations: what gets torn down, what gets carried, the repository layout, the technology choices, and the complete schema for Rooms 1–3.

Plan 2 covers research methodology — cost models, benchmarks, walk-forward design, multiple-testing control, and the seasonality validation layer. **Plan 3** covers execution — phase-by-phase steps, week-by-week schedule, what exists versus what must be built, risks, and the questions still open.

Nothing in this document is a promise that the research will find an edge. On the evidence measured during this audit, the most likely outcome is a rigorous negative result. That is stated here at the top so it cannot be read as a surprise later.

1. The audit findings that shaped this plan

Every number below was measured directly during this session, not taken from MICCV2's documentation.

1.1 What MICCV2 actually is

Verified	Result
Tests	**486 pass**, 18.5s
`verify_suite`	163 checks · 162 pass · 0 critical-fail · 1 warn
`verify_v3` (prod)	44/44 GREEN, 428s
Book CAGR (recomputed)	7.96% vs benchmark 8.89%
Warehouse	7,749,148 price rows · 174,616,363 F&O rows · 4,200 symbols · 2005-01-03 → 2026-08-14
Write-once ledgers	Real — SQLite triggers blocked both a test UPDATE and DELETE
Automation	launchd genuinely running; `last_success=2026-08-14`

The engineering works. The **strategy** proved nothing: zero promoted edges, 22 verdicts all KILL or CONTEXT, champion trailing-24m Sharpe 0.11, and only **24 live-forward days of which 9 were captured LIVE**. That is the finding that motivated this rebuild, and it is a research finding, not a software defect.

1.2 Defects found in MICCV2 that this rebuild must not inherit

These were found during the audit and are listed so the new architecture can design them out rather than rediscover them.

#	Defect	Design response in the new system
1	7 of 12 dashboard pages return HTTP 500 whenever any writer holds the DuckDB lock; `verify_dashboard` tests in-process on an idle machine and never sees it	Readers use an immutable snapshot, never the live file. Availability tested *under* a held writer lock
2	`verify_v3.py` defaults to `dev`, and in `dev` reports 2 critical failures	Env must be explicit; unset fails loudly. No silent default
3	A `dev`-env verification shells out and rebuilds the **prod** warehouse, including live network fetches	Verification is strictly read-only. Rebuild and verify are separate verbs
4	Two walk-forward implementations; the live one (`wf_threshold`) includes the current period's own forward return	One implementation, one call site, a test asserting the training window ends strictly before the decision date
5	Backtest gate and live gate read different source tables and disagree on **13 of 259 rebalances (5.0%)**; 11 flip the decision	Single source of truth per computed quantity; a test asserting research and production read the same table
6	Champion Sharpe never deflated, while every challenger is	The trial counter applies to everything, incumbent included
7	Empty `data/raw/nse|bse|fno|global|macro|shp` — daily fetches archive nothing	Raw archive is mandatory and hash-verified before any parse
8	166 warehouse views hard-code `/Users/satya_03/...` inside `read_parquet()`	All paths relative to a resolved repo root; a test asserts no absolute path appears in any view DDL
9	`db/app_state.sqlite` tracked in git despite docs saying otherwise	`.gitignore` verified by a test that greps `git ls-files`

1.3 The institutional premise — measured, and it fails

The owner's plan assumes disclosed institutional deals carry repeatable information. That was testable today against `v1seed.bulk_deals` (223,450 rows, 2006-01-02 → 2026-07-08) and `v1seed.block_deals` (12,430 rows).

Finding A — 54.8% of bulk deals are same-day round trips.

Client	Client-stock-days	Round trips	%
GRAVITON RESEARCH CAPITAL LLP	6,748	6,748	**100%**
HRTI PRIVATE LIMITED	2,968	2,968	**100%**
TOWER RESEARCH CAPITAL	1,165	1,165	**100%**
XTX MARKETS LLP	857	857	**100%**
All bulk deals	144,053	79,012	**54.8%**
All block deals	*11,962*	*80*	*0.7%*

The most active "institutions" in the dataset are high-frequency market makers whose bulk-deal disclosure is a mechanical consequence of crossing the 0.5% volume threshold intraday. They end the day flat. Block deals are clean.

Finding B — after filtering the churn, 30,771 directional buy events, entry at next-session open, market-relative to Nifty 500 TR:

Horizon	n	Mean rel.	Median rel.	Hit rate	Naive *t*
1 month	17,339	−0.12%	−1.55%	44.7%	−0.77
3 months	16,594	−0.22%	−4.81%	42.0%	−0.81
12 months	15,498	**+7.80%**	**−10.68%**	**43.0%**	+11.61

One and three months are flat. The twelve-month mean is positive only because of right skew — **the median event loses 10.68% to the market and 57% underperform**. The naive *t* of 11.61 ignores that 12-month windows overlap heavily. Corrected:

```
monthly cohort t        = +3.61
moving-block bootstrap = +9.67%  95% CI [−2.46%, +22.81%]   <- includes zero
P(mean > 0)             = 93.7%                             <- fails a 95% bar
```

Finding C — and that is measured on survivors. 13,928 of 30,771 events (45%) were dropped: 7,354 symbols absent from the price spine, 6,574 hitting a delisting or gap inside the window.

Finding D — the good news: the attrition is fixable, not fatal. The price spine *does* contain dead stocks — 1,497 of 4,200 symbols stopped trading before August 2026 (314 pre-2010, 667 in 2010-2019, 322 in 2020-2024, 194 in 2025-2026). So the 7,354 misses are **symbol-naming mismatches between the deal feed and the price spine, not survivorship**. Fixing the identity layer (§5) converts most of them into usable events. The 6,574 delisting cases are real information and will be carried as realized outcomes, not dropped.

> **FALSIFIED 2026-08-22. The second sentence of Finding D is wrong, and it was the justification for Phase 3 being the four-week highest-priority phase.**
>
> Measured against the rebuilt spine, plain symbol matching resolves **91.30%** of the 235,880 deal rows. Of the 19,130 rows (8.16%) that remain unresolved after rights entitlements are set aside:
>
> - **0 of 680** symbols are recoverable by ISIN aliasing
> - **1 of 680** is known to *any* identity master
>
> They are not naming mismatches. There is nothing to map them to. 93% first appear in 2018 or later, and the companies are largely absent from `bse_scrip_master` altogether, which points at NSE Emerge listings the price spine does not cover — though that explanation could not be confirmed from the data held, and the decision does not rest on it.
>
> The identity layer cannot convert these events because conversion needs a target that does not exist. They leave the universe on coverage grounds instead (decision 0032), and **Phase 3's headline justification goes with them** — it is still required for PIT sectors, participant classification and delisting handling, but the four-week estimate was built on recovering these events and should be re-cut.
>
> Left standing above rather than rewritten: the claim is what Phase 3 was sold on, and deleting it would hide why the phase was sized the way it was.

Consequence for scope. This is why the owner chose research-only with no engines. Engines A, B, E and G in the original plan are not being built. §2.2 records why.

1.4 Seasonality — already answered

The existing atlas in `db/v3_prod.duckdb` holds **31,893,556 cells** and already implements every dimension the plan's §11.2 specifies:

Plan asks for	Atlas has
13 windows (1…90)	✅ exactly those
4 alignment schemes	✅ CALENDAR_DATE 15,083,303 · TRADING_DAY_OF_YEAR 12,818,902 · MONTH_END 1,997,179 · MONTH_START 1,994,172
Price + market-relative	✅ 16,105,466 / 15,788,090
Stocks, indices, pooled	✅ 3,638 stocks · 46 indices · 1 pooled
Window-specific baseline	✅ `base_rate` per cell
Multiple-testing correction	✅ Benjamini–Yekutieli, 14,036 survive q<0.05
Permutation test	✅ White's Reality Check, 300 circular rotations

And its own verdict: **1,579,659 "significant" cells observed against 1,497,584 expected by chance — a ratio of 1.05.** The best NIFTY50 pattern sits at the **94th percentile of what randomly rotated data produces.**

Per owner decision Q42, the atlas is **rebuilt from scratch with new code**, not carried. The validation layer described in Plan 2 §7 is the genuinely new work.

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2. Scope

2.1 In scope for v1

Room	Contents	Status
Room 1	Raw archive (SHA-256, gzipped originals + parsed parquet), NSE + BSE bulk/block collectors, FII/DII collector, participant-OI proxy	Build
Room 2A	Individual deal outcomes across 9 horizons	Build
Room 2B	Institutional behaviour analysis (descriptive; no participant *ranking* without correction)	Build
Room 2C	Three deal interpretations — individual / accumulated / confirmation	Build
Room 3	Seasonality atlas rebuild + validation layer	Build
Room 6	Monitoring, data-quality gates, pause logic	Build (research-scope)

2.2 Explicitly out of scope for v1, with reasons

Deferred	Reason
Engine A (participant)	The most active directional buyer has 268 buys in 20 years; SBI MF has 80. Skill cannot be separated from luck at that sample size with 12-month overlapping horizons
Engine B (fund house)	Same power problem, plus parent-relationship data does not exist
Engine E (FII/DII regime)	`fii_dii_data` holds **22 days** (2026-06-17 → 2026-07-08). Cannot be researched until history accrues
Engine G (institutional × seasonality)	Combining two individually-null signals multiplies false-discovery risk. The single most dangerous item in the original plan
Room 5 (paper portfolios)	Owner decision Q6: no paper book until something passes gates
Engines C, D	Deferred, not rejected. The schemas are designed so they drop in (§7)

2.3 The plan's blind spot, and how this design closes it

The original plan applies rigorous false-discovery control to seasonality (§11) and **none at all** to participant selection (§12). Choosing "the best of 27,417 participants" is a larger multiple-testing problem than 31.9M seasonality cells, because each participant has far fewer observations.

Design response: the participant-ranking correction is specified in Plan 2 §6 *before* any leaderboard is computed, and `study_result` (§7.4) cannot store a participant-level claim without a populated `correction_method` and `corrected_p_value`. The schema makes the omission impossible rather than discouraged.

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3. Teardown and migration

Owner decision: full teardown, new repo, MICCV2 frozen read-only, carrying raw data and dropping derived data.

3.1 What is carried

Source	Size	Disposition
`MICCV2/data/raw/v1_export/`	1.2 GB, 126 parquet files	**Copy.** Irreplaceable — NSE does not serve 21 years of history. This is the seed for everything
`MICCV2/db/miccv2.duckdb`	274 KB	**Read-only reference** during migration; not carried into the new warehouse
`MICCV2/db/app_state.sqlite`	148 KB	**Extract only** the cumulative trial counter (N=47 + 21 legacy) per Q47
Trading-calendar logic	code	**Port**, do not re-derive. 5,339 observed sessions, verified complete
Cost-model constants	config	**Port as the floor**, then extend per Plan 2 §4

3.2 What is dropped

Dropped	Size	Why
`MICCV2/data/warehouse/`	1.5 GB	Derived. Rebuilt by new code from `v1_export`
`MICCV2/data/v3/`	2.3 GB	Derived, including the seasonality atlas (rebuilt per Q42)
All `src/` strategy code	~27k lines	Champion, GCE, engines, ML tracks, factory — the research paths that proved nothing
`src/dashboard/`	12 pages	Deferred to Phase 8 per Q51
Docker / CI image work	—	Skipped per Q52

3.3 Freezing MICCV2 — exact steps

Per Q50, all analysing agents are removed. This is the only step in this plan that touches the existing machine, and it is reversible.

```
# 1. Unload the launchd agents (stops all writes to carried data)
launchctl bootout gui/$(id -u)/com.miccv2.daily
launchctl bootout gui/$(id -u)/com.miccv2.dashboard
launchctl bootout gui/$(id -u)/com.miccv3.heartbeat

# 2. Move the plists out of the load path (kept, not deleted)
mkdir -p ~/Workspace/MICCV2/_frozen_agents
mv ~/Library/LaunchAgents/com.miccv2.daily.plist      ~/Workspace/MICCV2/_frozen_agents/
mv ~/Library/LaunchAgents/com.miccv2.dashboard.plist  ~/Workspace/MICCV2/_frozen_agents/
mv ~/Library/LaunchAgents/com.miccv3.heartbeat.plist  ~/Workspace/MICCV2/_frozen_agents/

# 3. Tag the final state, so the 107-commit history stays addressable
cd ~/Workspace/MICCV2 && git tag -a frozen-2026-08-16 -m "final state before rebuild"

# 4. Verify nothing is still running
launchctl list | grep -i micc     # expect: no output
```

Not done: deleting MICCV2 from disk. It stays as a read-only reference until the new system reproduces the warehouse and the owner says otherwise.

3.4 Migration verification gate

The new warehouse is not accepted until it reproduces MICCV2's numbers from the same seed. This is the one place the old system is genuinely useful — as an oracle.

Check	Expected
Price rows	7,749,148
Distinct symbols	4,200
Date span	2005-01-03 → 2026-08-14
F&O rows	174,616,363
Bulk deals	223,450
Block deals	12,430
Trading sessions	5,339
Dead symbols (last trade < 2026-08-01)	1,497

A mismatch on any row is a blocking failure, investigated before proceeding.

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4. Repository layout and stack

4.1 Stack decision

Python 3.14 · DuckDB · Parquet · SQLite · pytest. Rationale, given the owner's "job level" framing:

What reads as senior in a quant portfolio, in order: point-in-time correctness · reproducible deterministic builds · pre-registration and honest negative results · tests that pin behaviour · clean module boundaries. Database choice is far down that list. A reviewer will try to *clone and run* the project first — a zero-server stack wins there.

Layer	Choice	Why
Immutable raw archive	Parquet + gzipped originals, SHA-256	Plan §5. Append-only, never rewritten
Analytics + relational marts	DuckDB	174M rows already scan fine on this laptop. Arrow-native, zero-server, the current research standard
Append-only ledgers	SQLite	Only because it has real triggers. MICCV2's write-once enforcement genuinely works — verified
Review queue / mutable state	SQLite	27,417 names is small; needs transactions, not analytics
Migrations	Plain SQL files + `schema_migrations` table	No ORM. MICCV2's V2 store drifted to 1 table vs prod's 15 precisely because it had none

Rejected: Postgres. It adds a running service, connection management, and a migration of 7.7M + 174M rows, for no capability this workload needs. A sharp interviewer would ask why, and there is no good answer at this scale.

4.2 Directory layout

```
institutional-research/
├── README.md                  honest status, leads with what is NOT working
├── pyproject.toml             uv-managed, Python 3.14 pinned
├── configs/
│   ├── sources.yml            every source: URL, cadence, licence, contact
│   ├── costs.yml              the advanced cost model (Plan 2 §4)
│   ├── benchmarks.yml         the 6-benchmark set (Plan 2 §5)
│   ├── universe.yml           liquidity and eligibility rules
│   ├── participants.yml       classification patterns + behavioural thresholds
│   └── research.yml           horizons, CPCV folds, correction methods
├── migrations/
│   └── 0001_init.sql …        forward-only, checksummed, never edited
├── src/
│   ├── common/                paths, config, logging, calendar, hashing
│   ├── archive/               Room 1 — fetch, hash, store, verify
│   ├── ingest/                parsers: nse_bulk, nse_block, bse_bulk, bse_block, fii_dii
│   ├── identity/              security master, symbol_history, participants
│   ├── warehouse/             spine, universe, benchmarks, PIT sectors
│   ├── mart/                  institutional_deals_clean, interpretations
│   ├── research/              outcomes, cost model, CPCV, corrections
│   ├── seasonality/           atlas build + validation layer
│   ├── governance/            experiment registry, provenance DAG, trial counter
│   └── monitor/               Room 6 — DQ gates, pause logic, alerts
├── tests/                     mirrors src/ one-to-one
├── scripts/                   one-shot tools, report builder, launchd plists
├── docs/
│   ├── plan/                  these three documents
│   ├── decisions/             one file per owner decision, dated and attributed
│   └── reports/               generated markdown + PDF
├── data/                      gitignored
│   ├── raw/                   immutable archive, never rewritten
│   ├── warehouse/             parquet marts, regenerable
│   └── snapshots/             read-only DB copies for readers
└── db/                        gitignored
    ├── research.duckdb        analytics + marts
    ├── governance.sqlite      append-only ledgers
    └── review.sqlite          participant review queue
```

4.3 Standing engineering rules

Carried from MICCV2 where they earned their place, plus new ones from §1.2.

1. **No order-placement code.** Not in tests, not commented out.
2. **No live trading.** Ever, in this repo.
3. **Readers never touch a live database file.** They read `data/snapshots/`.
4. **Verification is read-only.** A verify command that writes is a bug.
5. **The environment must be explicit.** Unset `RESEARCH_ENV` fails loudly.
6. **No absolute paths in any stored view DDL.** Tested.
7. **A fix is done when its test has been watched failing**, not when the code changes.
8. **UNKNOWN beats inference.** Where something cannot be established, it says so.
9. **Raw files are never overwritten or deleted.**
10. **Every study is pre-registered.** No exceptions (Q44).

5. Room 1 — Raw archive and collection

5.1 Design

Owner decisions: both daily and historical endpoints (Q7) · backfill the 2026-07-08 → present gap first (Q8) · collect BSE too (Q9) · browser-like session with rate limiting (Q10) · store original bytes gzipped *and* parsed parquet (Q11) · retain forever (Q12) · automated from day one (Q13).

```
fetch  ──►  bytes in memory
              │
              ├──► SHA-256 ──► if hash already in deal_source_files: STOP (idempotent)
              │
              ├──► write raw/institutional_deals/<ex>/<type>/year=YYYY/month=MM/<name>.gz
              │
              ├──► parse ──► write the same path as .parquet
              │
              └──► INSERT deal_source_files (status=OK | PARSE_FAILED | FETCH_FAILED)
```

A parse failure **still archives the bytes** and records the failure. That is the difference between a system that can recover from a source format change and one that silently loses a day.

5.2 `deal_source_files`

```
CREATE TABLE deal_source_files (
    source_file_id   BIGINT PRIMARY KEY,
    exchange         TEXT    NOT NULL,      -- NSE | BSE
    report_type      TEXT    NOT NULL,      -- BULK | BLOCK | FII_DII | SHP
    source_url       TEXT    NOT NULL,
    report_date      DATE    NOT NULL,
    downloaded_at    TIMESTAMP NOT NULL,
    file_name        TEXT    NOT NULL,
    file_hash        TEXT    NOT NULL,      -- SHA-256 of the raw bytes
    file_bytes       BIGINT  NOT NULL,
    parser_version   TEXT    NOT NULL,
    row_count        INTEGER,
    ingestion_status TEXT    NOT NULL,      -- OK|PARSE_FAILED|FETCH_FAILED|EMPTY
    error_message    TEXT,
    http_status      INTEGER,
    fetch_duration_ms INTEGER,
    UNIQUE (file_hash),
    UNIQUE (exchange, report_type, report_date, parser_version)
);
```

`UNIQUE(file_hash)` makes re-fetching free and makes "did the source change?" answerable — a new hash for a date already collected is a **revision**, which §5.4 handles explicitly.

5.3 `institutional_deals_raw`

Verbatim source rows. Never cleaned, never corrected.

```
CREATE TABLE institutional_deals_raw (
    raw_deal_id      BIGINT PRIMARY KEY,
    source_file_id   BIGINT NOT NULL REFERENCES deal_source_files,
    exchange         TEXT   NOT NULL,
    deal_type        TEXT   NOT NULL,       -- BULK | BLOCK
    trade_date       DATE   NOT NULL,
    symbol_raw       TEXT,
    security_name_raw TEXT,
    client_name_raw  TEXT,
    side_raw         TEXT,
    quantity_raw     TEXT,                  -- TEXT: source sometimes has commas
    deal_price_raw   TEXT,
    remarks_raw      TEXT,
    raw_row_json     TEXT   NOT NULL,       -- the complete original row
    row_index        INTEGER NOT NULL,      -- position within the file
    ingested_at      TIMESTAMP NOT NULL
);
```

5.4 Source revisions

NSE and BSE silently restate. A revision is a new `file_hash` for a `(exchange, report_type, report_date)` already held.

```
CREATE TABLE source_revisions (
    revision_id      BIGINT PRIMARY KEY,
    exchange         TEXT NOT NULL,
    report_type      TEXT NOT NULL,
    report_date      DATE NOT NULL,
    prior_file_id    BIGINT NOT NULL REFERENCES deal_source_files,
    new_file_id      BIGINT NOT NULL REFERENCES deal_source_files,
    detected_at      TIMESTAMP NOT NULL,
    rows_added       INTEGER,
    rows_removed     INTEGER,
    rows_changed     INTEGER,
    review_status    TEXT NOT NULL DEFAULT 'PENDING'
);
```

Both versions are kept. Research uses the version available at the decision date — a restatement published in 2020 cannot inform a 2015 signal. This is the point-in-time discipline applied to the source layer itself, and it is a thing MICCV2 never had.

5.5 Collection schedule

Source	Cadence	Window	Notes
NSE bulk	daily 19:00 IST	T	Published after close
NSE block	daily 19:00 IST	T	
BSE bulk	daily 19:15 IST	T	New — never collected before
BSE block	daily 19:15 IST	T	New
FII/DII	daily 19:30 IST	T	Starts accruing now (Q4 route: both)
SHP / promoter	quarterly	filing dates	For promoter list (Q25)

Rate limit 1 request / 2 s, honest User-Agent, exponential backoff, and **the retry never re-fetches a hash already held**.

---

6. Identity layer

This is the highest-priority component (Q16) because it directly determines whether the outcome study is biased. 7,354 of 30,771 events currently fail to resolve, and §1.3 Finding D established these are naming mismatches, not delistings.

6.1 `security_master` — wraps the existing masters (Q15)

```
CREATE TABLE security_master (
    security_id      BIGINT PRIMARY KEY,
    isin             TEXT UNIQUE,
    canonical_symbol TEXT NOT NULL,
    company_name     TEXT NOT NULL,
    listing_date     DATE,
    delisting_date   DATE,                  -- populated from last-trade detection
    delisting_reason TEXT,                  -- MERGER|ACQUISITION|SUSPENSION|UNKNOWN
    status           TEXT NOT NULL,         -- ACTIVE|DELISTED|SUSPENDED|MERGED
    merged_into_id   BIGINT REFERENCES security_master,
    source           TEXT NOT NULL,
    confidence       TEXT NOT NULL          -- HIGH|MEDIUM|LOW
);
```

`merged_into_id` matters: when a stock merges, its holder receives shares in the acquirer. Treating that as a delisting-to-zero is a large downward bias; ignoring it is an upward one. Both are wrong and MICCV2 did the second.

6.2 `symbol_history` — the point-in-time map

```
CREATE TABLE symbol_history (
    symbol_history_id BIGINT PRIMARY KEY,
    security_id      BIGINT NOT NULL REFERENCES security_master,
    symbol           TEXT NOT NULL,
    exchange         TEXT NOT NULL,
    series           TEXT,
    valid_from       DATE NOT NULL,
    valid_to         DATE,                  -- NULL = current
    source           TEXT NOT NULL,
    UNIQUE (symbol, exchange, valid_from)
);
```

Resolution rule, enforced in one function and one only:

```
def resolve(symbol: str, exchange: str, on_date: date) -> SecurityId | Unresolved:
    """The symbol identity valid ON the transaction date — never today's."""
```

Every caller uses it. A test asserts no module builds its own symbol lookup.

6.3 `sector_history` — point-in-time sectors (Q31)

The owner chose to build PIT sectors. MICCV2's `dim_sector` is current-only, which is a look-ahead: a company classified "IT" today may have been "Textiles" in 2008.

```
CREATE TABLE sector_history (
    security_id      BIGINT NOT NULL REFERENCES security_master,
    sector           TEXT NOT NULL,
    industry         TEXT,
    classification   TEXT NOT NULL,         -- NIC|NSE|BSE|MANUAL
    valid_from       DATE NOT NULL,
    valid_to         DATE,
    confidence       TEXT NOT NULL,
    PRIMARY KEY (security_id, classification, valid_from)
);
```

Sources, in preference order: NSE index-membership history (`index_membership` has 13,163 rows with `effective_from` / `effective_to`), BSE scrip master vintages, then manual. Where no vintage exists, the earliest known classification is carried backward and marked `confidence='LOW'` — and the outcome study reports sector-relative results **separately** for LOW-confidence rows so the reader can discount them.

6.4 Participant identity

Owner decisions: exact-after-cleaning, no fuzzy auto-merge (Q19) · variants stay separate until the owner rules (Q20) · manual mapping file for fund houses (Q21) · HFT flagged as a category (Q18).

```
CREATE TABLE participant_master (
    participant_id   BIGINT PRIMARY KEY,
    canonical_name   TEXT NOT NULL UNIQUE,
    participant_type TEXT NOT NULL,         -- see §6.5
    classification_method TEXT NOT NULL,    -- BEHAVIOURAL|NAME_PATTERN|MANUAL
    parent_group_id  BIGINT REFERENCES participant_master,
    country          TEXT,
    confidence_level TEXT NOT NULL,         -- HIGH|MEDIUM|LOW|UNKNOWN
    first_seen       DATE,
    last_seen        DATE,
    deal_count       INTEGER,
    review_status    TEXT NOT NULL,         -- PENDING|REVIEWED|AUTO
    reviewed_by      TEXT,
    reviewed_at      TIMESTAMP,
    review_notes     TEXT
);

CREATE TABLE participant_aliases (
    alias_id         BIGINT PRIMARY KEY,
    participant_id   BIGINT NOT NULL REFERENCES participant_master,
    raw_name         TEXT NOT NULL UNIQUE,
    normalized_name  TEXT NOT NULL,
    mapping_method   TEXT NOT NULL,         -- EXACT|CLEANED|MANUAL
    mapping_confidence TEXT NOT NULL,
    suggested_merge_id BIGINT REFERENCES participant_master,  -- surfaced, never applied
    review_status    TEXT NOT NULL,
    reviewed_at      TIMESTAMP,
    review_notes     TEXT
);
```

`suggested_merge_id` is how Q20 is honoured: `QE SECURITIES LLP` and `QE SECURITIES` get a suggestion recorded, stay separate entities, and the owner decides later. Nothing merges automatically.

6.5 Classification — behavioural first, then name

The audit established that the best classifier for the largest category is **behaviour, not name**. This is the single most useful thing found.

Category	Rule	Names	% of deals
`PROP_HFT`	≥20 client-stock-days AND ≥95% same-day round-trip	327	**44.0%**
`INDIVIDUAL`	name matches personal-name pattern	12,891	22.9%
`BROKER_SEC`	`SECURITIES|BROKING|BROKERS|STOCK`	947	7.3%
`FPI_OFFSHORE`	`MAURITIUS|SINGAPORE|CYPRUS|CAYMAN|GLOBAL MARKETS`	852	1.9%
`MUTUAL_FUND`	`MUTUAL FUND|ASSET MANAGE|AMC`	806	1.1%
`BANK`	`\bBANK\b`	235	0.5%
`INSURANCE`	`INSURANCE|ASSURANCE`	137	0.2%
`PENSION_SOVEREIGN`	`PENSION|SOVEREIGN|MONETARY AUTH`	32	0.0%
`UNKNOWN`	everything else	11,190	22.2%

Manual review queue: 1,515 names — those UNKNOWN with ≥6 deals. This is the better answer Q17 invited: classification is 78% mechanical, and the mechanical part for the biggest category is behavioural, which is far more defensible than a hand-maintained name list. Graviton never declares itself a market maker; 6,748 of 6,748 round-trips does.

Ordering matters: **behavioural runs first.** A firm named `… SECURITIES PRIVATE LIMITED` that round-trips 100% of the time is `PROP_HFT`, not a broker. Name patterns only see what behaviour did not already classify.

Where the 95% / 20 thresholds come from, and how they are defended. They were not tuned. The observed distribution is close to bimodal: the top twenty participants by activity sit at 100.0%, 100.0%, 100.0%, 98.9%, 85.6%, 62.0%, 52.9% … — a dense cluster at ~100% and a long tail below 90%. 95% sits inside the gap; 20 client-stock-days is the point at which the round-trip ratio stops being dominated by small-sample noise.

Both are configuration, and **both get a sensitivity table in every study**:

```
prop_hft_classifier:
  roundtrip_ratio: 0.95       # sensitivity: 0.80, 0.90, 0.99
  min_client_stock_days: 20   # sensitivity: 10, 50
```

The reported N of eligible events at each of the nine combinations is published alongside the headline. A finding that depends on whether the cut is 0.90 or 0.95 is a finding about the cut, not about institutions — and this is a 44%-of- data filter, so that check is not optional.

6.5.1 Review queue ordering

The 1,515-name queue is worked in descending order of **`deal_value × current contribution to the unresolved-symbol rate`**, not raw deal count. That front- loads the names which most affect the Phase 3 gate (unresolved rate < 5%), so the gate can pass before the queue is exhausted. Progress against the gate is printed after every batch.

6.6 Promoter list (Q25)

No promoter list exists. Built from shareholding-pattern (SHP) filings, which disclose promoter-group entity names per company per quarter.

```
CREATE TABLE promoter_entities (
    security_id      BIGINT NOT NULL REFERENCES security_master,
    entity_name      TEXT NOT NULL,
    normalized_name  TEXT NOT NULL,
    valid_from       DATE NOT NULL,         -- the quarter's filing date
    valid_to         DATE,
    holding_pct      REAL,
    source_file_id   BIGINT REFERENCES deal_source_files,
    PRIMARY KEY (security_id, normalized_name, valid_from)
);
```

A deal is `promoter_related_flag = TRUE` when the participant's normalized name matches a promoter entity of that security **valid on the trade date** — a point-in-time join, not a current-list lookup.

---

7. Room 2 — the clean mart and research schemas

7.1 `institutional_deals_clean`

Owner decisions: keep NSE/BSE duplicates with a group id (Q24) · add a 5-day round-trip flag alongside the same-day one (Q23) · treat a client's multiple same-day rows as **separate** events (Q28) · exclude `PROP_HFT` from `eligible_for_research` by default but keep the rows (Q27) · size filter `deal_value_to_adv20 ≥ 0.5%` and ≥ ₹1cr, configurable (Q26).

```sql
CREATE TABLE institutional_deals_clean (
    deal_id          BIGINT PRIMARY KEY,
    raw_deal_id      BIGINT NOT NULL REFERENCES institutional_deals_raw,
    security_id      BIGINT REFERENCES security_master,
    participant_id   BIGINT REFERENCES participant_master,
    trade_date       DATE NOT NULL,
    available_from   TIMESTAMP NOT NULL,    -- when the disclosure was observable
    entry_date       DATE NOT NULL,         -- first tradable session after that
    exchange         TEXT NOT NULL,
    deal_type        TEXT NOT NULL,
    side             TEXT NOT NULL,         -- BUY | SELL
    quantity         BIGINT NOT NULL,
    deal_price       REAL NOT NULL,
    gross_deal_value REAL NOT NULL,
    adv20            REAL,
    deal_value_to_adv20 REAL,

    duplicate_group_id      BIGINT,
    same_day_round_trip_flag BOOLEAN NOT NULL,
    five_day_round_trip_flag BOOLEAN NOT NULL,
    internal_transfer_flag  BOOLEAN NOT NULL,
    promoter_related_flag   BOOLEAN NOT NULL,
    suspect_flag            BOOLEAN NOT NULL,
    unresolved_symbol_flag  BOOLEAN NOT NULL,
    eligible_for_research   BOOLEAN NOT NULL,
    ineligibility_reason    TEXT,

    clean_version    TEXT NOT NULL,
    created_at       TIMESTAMP NOT NULL
);
```

`available_from` is the field the whole study rests on. It is a timestamp, not a date, and it is established empirically in Phase 1 by recording the observed publication time of each report over several weeks — never assumed. If publication time cannot be established for a historical period, `available_from` is set to the conservative bound (next session open) and the row carries `confidence='LOW'`.

7.2 The three interpretations (plan §10)

Kept strictly separate, never silently mixed.

```sql
CREATE TABLE deal_interpretation (
    interpretation_id BIGINT PRIMARY KEY,
    mode             TEXT NOT NULL,      -- INDIVIDUAL|ACCUMULATED|CONFIRMATION
    deal_id          BIGINT NOT NULL REFERENCES institutional_deals_clean,
    position_id      BIGINT,             -- groups an accumulation sequence
    sequence_index   INTEGER,            -- 1 = initiating deal
    is_initiation    BOOLEAN NOT NULL,
    is_confirmation  BOOLEAN NOT NULL,
    days_since_prior INTEGER,
    cumulative_qty   BIGINT,
    cumulative_value REAL,
    version          TEXT NOT NULL
);
```

Accumulation grouping rule: same `participant_id` + `security_id`, same side, gaps ≤ N trading days (configurable, default 63). A gap longer than N closes the position and a later deal starts a new one.

7.3 `deal_forward_outcomes`

Horizons: **1/2/3/5/10/21 trading sessions** primary, 3/6/12 months robustness — live values in `configs/research.yml`, rationale in decision 0004. This supersedes the nine-month-grid answer to Q29, which measured power showed was pointed at a frequency where MDE exceeds any plausible effect.

Other owner decisions here: delisted stocks used, not dropped (Q32) · five-plus benchmarks (Q30) · intraday excursions primary with close reported alongside (Q33).

```
CREATE TABLE deal_forward_outcomes (
    outcome_id        BIGINT PRIMARY KEY,
    deal_id           BIGINT NOT NULL REFERENCES institutional_deals_clean,
    interpretation_id BIGINT REFERENCES deal_interpretation,
    horizon_months    INTEGER NOT NULL,      -- 1,3,6,8,10,12,15,18,24
    entry_date        DATE NOT NULL,
    entry_price       REAL NOT NULL,
    exit_date         DATE NOT NULL,
    exit_price        REAL NOT NULL,
    exit_reason       TEXT NOT NULL,         -- HORIZON|DELISTED|MERGED|SUSPENDED

    stock_return      REAL NOT NULL,
    net_return        REAL NOT NULL,         -- after the Plan 2 §4 cost model

    -- one row per benchmark, see benchmark_returns below
    max_adverse_excursion   REAL,
    max_favorable_excursion REAL,
    days_to_max_adverse     INTEGER,
    excursion_basis         TEXT NOT NULL, -- CLOSE|INTRADAY

    outcome_complete_flag BOOLEAN NOT NULL,
    calculation_version   TEXT NOT NULL,
    UNIQUE (deal_id, interpretation_id, horizon_months, calculation_version)
);

CREATE TABLE outcome_benchmark_returns (
    outcome_id        BIGINT NOT NULL REFERENCES deal_forward_outcomes,
    benchmark_id      TEXT NOT NULL,
    benchmark_return REAL NOT NULL,
    relative_return  REAL NOT NULL,
    PRIMARY KEY (outcome_id, benchmark_id)
);
```

Splitting benchmarks into their own table is what makes Q30's "at least 5 benchmarks" workable without nine near-identical columns, and lets a benchmark be added later without a schema change.

Delisting handling (Q32). `exit_reason` distinguishes four cases and each is priced differently: `HORIZON` normal · `DELISTED` exit at last traded price with a configurable recovery haircut · `MERGED` roll into `merged_into_id` at the exchange ratio and continue the horizon · `SUSPENDED` mark to last price and flag. The choice materially moves the headline number, so all four are reported separately and the aggregate is shown under each assumption.

7.4 `study_result` — replaces the signal ledger for v1

This is the answer to owner question Q2. The full reasoning is in Plan 3 §1; the short version: the *experiment registry* is required now, the *signal ledger* is not, because a signal ledger records engine decisions and there are no engines. What research needs instead is a result store.

```
CREATE TABLE study_result (
    result_id        BIGINT PRIMARY KEY,
    experiment_id    TEXT NOT NULL REFERENCES experiment_registry,
    stratum          TEXT NOT NULL,        -- 'ALL' | 'sector=IT' | 'participant=SBI_MF'
    stratum_type     TEXT NOT NULL,
    horizon_months   INTEGER,
    benchmark_id     TEXT,
    n_events         INTEGER NOT NULL,
    n_independent    INTEGER NOT NULL,     -- after overlap adjustment
    mean_return      REAL,
    median_return    REAL,
    hit_rate         REAL,
    raw_p_value      REAL,
    corrected_p_value REAL,
    correction_method TEXT NOT NULL,       -- NOT NULL: the §2.3 guard
    n_tests_in_family INTEGER NOT NULL,
    bootstrap_ci_low  REAL,
    bootstrap_ci_high REAL,
    verdict          TEXT NOT NULL,        -- PASS|FAIL|UNDERPOWERED
    input_hashes     TEXT NOT NULL,        -- provenance DAG, Plan 2 §8
    code_commit      TEXT NOT NULL,
    computed_at      TIMESTAMP NOT NULL
);
```

`correction_method` and `n_tests_in_family` are `NOT NULL`. A participant-level claim cannot physically be stored without declaring how many participants were tested. That is §2.3's blind spot closed in the schema.

7.5 Engine schemas — designed now, unpopulated (Q1)

Per owner decision Q1, engine tables are created so engines drop in later without a migration that rewrites research history.

```sql
CREATE TABLE engine_config (
    engine_id        TEXT PRIMARY KEY,
    engine_name      TEXT NOT NULL,
    purpose          TEXT NOT NULL,
    data_inputs      TEXT NOT NULL,
    participant_level TEXT NOT NULL,
    allowed_sides    TEXT NOT NULL,
    interpretation_mode TEXT NOT NULL,
    holding_periods  TEXT NOT NULL,
    entry_policy     TEXT NOT NULL,
    exit_policy      TEXT NOT NULL,
    liquidity_policy TEXT NOT NULL,
    risk_policy      TEXT NOT NULL,
    benchmark_policy TEXT NOT NULL,
    minimum_history_policy TEXT NOT NULL,
    false_discovery_policy TEXT NOT NULL,   -- required before an engine may run
    enabled_status   TEXT NOT NULL DEFAULT 'DISABLED',
    version          TEXT NOT NULL
);

CREATE TABLE institutional_signal_ledger (
    signal_id        BIGINT PRIMARY KEY,
    engine_id        TEXT NOT NULL REFERENCES engine_config,
    as_of_date       DATE NOT NULL,
    deal_id          BIGINT REFERENCES institutional_deals_clean,
    security_id      BIGINT REFERENCES security_master,
    participant_id   BIGINT REFERENCES participant_master,
    interpretation_mode TEXT NOT NULL,
    intended_horizon INTEGER,
    signal_type      TEXT NOT NULL,
    signal_status    TEXT NOT NULL,        -- APPROVED|REJECTED|BLOCKED|SKIPPED
    reason           TEXT NOT NULL,
    seasonality_cell_id BIGINT,
    engine_config_version TEXT NOT NULL,
    experiment_id    TEXT REFERENCES experiment_registry,
    input_hashes     TEXT NOT NULL,
    code_commit      TEXT NOT NULL,
    created_at       TIMESTAMP NOT NULL
);
```

Both tables are created empty in migration `0001`, with a test asserting they stay empty while `enabled_status='DISABLED'` for every engine. When engines are built, the schema does not move.

---

8. What exists, what must be built

Component	Status	Source
21-year price history	✅ Have	`v1_export`, 7.7M rows, 4,200 symbols incl. 1,497 dead
F&O history	✅ Have	174.6M rows, 2005-2026
Bulk deals	✅ Have to 2026-07-08	223,450 rows; needs 5-week backfill
Block deals	✅ Have to 2026-07-08	12,430 rows
Trading calendar	✅ Have	5,339 sessions, verified complete
ISIN master + renames	🟡 Partial	276 rename rows — insufficient for 7,354 misses
Index history	🟡 Partial	NIFTY 50/100/200/Midcap100 good; **Smallcap has 15 rows**
Nifty 500 TR benchmark	✅ Have	cap-weighted + dividends, 2011-2026
Participant OI (FII/DII proxy)	✅ Have	15,359 rows, 2014-2026
FII/DII cash	❌ 22 days	Must collect forward
BSE bulk/block	❌ None	Must build
Raw archive	❌ None	Must build — dirs exist and are empty
PIT sectors	❌ None	Must build from `index_membership`
Promoter list	❌ None	Must build from SHP
Participant identity	❌ None	Must build; 78% automatable
Seasonality atlas	🔄 Rebuild	31.9M cells exist; rebuilding per Q42

The three genuine gaps that need outside input: a smallcap index history (Plan 2 §5 proposes constructing one), FII/DII cash history (unobtainable — accrues forward only), and the publication-time evidence for `available_from` (measured over the first weeks of collection).

Continues in Plan 2 — Research Methodology and Plan 3 — Execution.